

PROJECT REPOSITORY

github.com/vishal-
cse185/edupulse-academic-
platform

PRIMARY FRAMEWORK

Flutter 3.7+ & Dart 3.0+

AUTOMATED QA STATUS

100% Test Pass Rate (5/5 Suites)

1. Executive Summary

EduPulse is an enterprise-grade academic management system designed to streamline institutional governance, faculty workflows, and student success tracking. Built on a clean, decoupled architecture, it offers continuous attendance compliance tracking, automated rubric grading, multi-factor risk diagnostics, interactive GPA simulation, and official academic audit transcripts.

2. Technology Stack Breakdown

LAYER / DOMAIN	TECHNOLOGY / TOOL	KEY RESPONSIBILITIES & IMPLEMENTATION
Frontend Client	Flutter 3.7+ (Web, Desktop, Mobile)	Single codebase multi-platform UI with responsive layouts, Material 3, and zero flex overflows.
Core Language	Dart 3.0+	Type-safe, null-safe language utilizing sound static typing, isolates, and native performance.
State Management	Provider Architecture	Reactive <code>ChangeNotifier</code> listeners, scoped models, and decoupled dependency injection.
Design & Typography	Google Fonts (Inter) & Slate/Indigo Palette	Executive enterprise design tokens, high contrast legibility, and dynamic micro-trend indicators.
Diagnostic Engine	Adaptive Academic Intelligence Service	Multi-factor continuous risk evaluation, concept bottleneck diagnosis, and remediation checklist generator.
Testing & QA	Flutter Test / Dart Unit & Widget Framework	5 test suites covering models, algorithms, widgets, workflows, and navigation with 100% pass rate.
DevOps & CI/CD	GitHub Actions Workflow (<code>ci.yml</code>)	Automated static linting, test execution, and production web compilation on every commit.
Containerization	Docker (Multi-stage) & Nginx Alpine	Production container with optimized tree-shaken asset delivery and Docker Compose

LAYER / DOMAIN	TECHNOLOGY / TOOL	KEY RESPONSIBILITIES & IMPLEMENTATION
		orchestration.

3. Academic Intelligence Mathematical Formulation

The platform computes real-time academic composite scores using weighted multi-factor diagnostic equations:

Composite Score = (0.40 × Exam Average) + (0.35 × Assignment Average) + (0.25 × Attendance Rate)



Compliance & Risk Heuristics

- **Low Risk (≥75%):** Good academic standing.
- **Medium Risk (60% - 74%):** Subject gaps flagged for remediation.
- **High Risk (<60% or Attendance <75%):** Automatic counselor alert dispatch.



Grade Calculation Engine

- Standard 4.0 GPA calculation algorithm.
- Continuous evaluation with automated preliminary rubric feedback.
- "What-If" GPA recovery projections.

4. Architecture & Security Standards



Layered Decoupled Architecture

Strict separation across Presentation (/features), Business Logic (/services), Data Models (/models), and Core Utilities (/core).



Institutional Governance & Audit

Official academic audit transcripts stamped with Dean of Academic Affairs signature line and digital SHA-256 verification hash.